

Class 10 Science - Chapter 7

How do Organisms Reproduce? (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)

Class 7-8 ki yeh cheezein pehle pakki karo, warna Ch7 samajh nahi aayega.

1. Reproduction kya hai (Class 7 base)

Reproduction = jeevon ka apne jaise naye jeev paida karna (apni species ko aage badhana). Yeh species ki nirantarta (continuity) ke liye zaroori. Reproduction ek aisa life process hai jo INDIVIDUAL ke zinda rehne ke liye zaroori NAHI, par SPECIES ke survival ke liye zaroori.

2. Cell aur Cell division (Class 8-9 base)

Cell = jeevan ki ikai. Naye cell purane cell ke baant-ne (division) se bante.
DNA = cell ke nucleus me, jeev ki "blueprint" (instruction) wala chemical.
Chromosome = DNA + protein se bana dhaaga jaisa structure (nucleus me).

3. Plant ke bhaag (Class 6-7 base)

Flower (phool) = plant ka reproductive (prajanan) bhaag.
Phool ke 4 main bhaag: Sepal (hara, bahari), Petal (rangeen pankhudi),
Stamen (pollen wala - male), Carpel/Pistil (female).
Seed (beej) se naya paudha; fruit (phal) seed ko dhakta+failata.

4. Male/Female ka idea (Class 8)

Gamete = reproductive cell (male = sperm/pollen, female = egg/ovum).
Fertilisation = male gamete + female gamete ka milan -> zygote.
Zygote = pehli cell jisse naya jeev banta.

5. Variation (Ch8 link)

Variation = ek hi species ke jeevon me thoda-thoda antar.
Sexual reproduction me variation jyada (do parents ka DNA mix);
asexual me bahut kam (ek hi parent ki copy).

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT

Class 10 Science | Ch 7: How do Organisms Reproduce?

[GREEN = board exam me baar-baar poocha jaata hai]

1. Reproduction aur DNA copying

Reproduction me DNA ki copy banti hai (taaki naye jeev ko blueprint mile).

DNA copying me chhoti galtiyan (errors) = VARIATION. Yeh variation evolution aur species ke survival (badalte environment me) ke liye useful.

2. Asexual Reproduction (ek parent) [TOP EXAM AREA]

Asexual = sirf EK parent, gamete ki zaroorat NAHI, offspring parent ki COPY (clone). Variation bahut kam.

Types (har ek ka example RATTA karo):

(1) FISSION:

- Binary fission: ek cell do me (Amoeba, Leishmania, bacteria).
- Multiple fission: ek cell se bahut (Plasmodium - malaria parasite).

(2) BUDDING: parent par "kali" (bud) nikalti jo alag hokar naya jeev (Hydra, Yeast).

(3) FRAGMENTATION: body tukdon me toot kar har tukda naya jeev (Spirogyra).

(4) REGENERATION: kata hua tukda poora jeev ban jaata (Planaria, Hydra).

(Note: regeneration = reproduction nahi hamesha; specialised cells se.)

(5) VEGETATIVE PROPAGATION (plants): jad/tana/patti se naya paudha.

- Tana: aloo (potato eyes), ginger.
- Patti: Bryophyllum (patti ke kinare se naye paudhe).
- Artificial: cutting, layering, grafting, tissue culture.

Faayda: tezi se, flower jaisa hi (same quality), jinme seed nahi (kela, gulab) unko bhi uga sakte.

(6) SPORE FORMATION: spore (chhote beej jaise) se naya jeev (Rhizopus - bread mould; thaili (sporangia) me spore bante).

3. Sexual Reproduction in Plants (Flower) [VERY IMPORTANT]

Phool ke 4 bhaag (bahar se andar):

- Sepal (hara, kali ko bachata) -> Petal (rangeen, insect attract)
- STAMEN (male) = Anther (pollen banata) + Filament (danthal).
- CARPEL/PISTIL (female) = Stigma (pollen pakadta) + Style (nali) + OVARY (andar OVULE/female gamete).

Bisexual flower = stamen+carpel dono (hibiscus); Unisexual = ek hi (papaya, watermelon).

POLLINATION = pollen ka anther se stigma tak pahunchna.

- Self-pollination: usi phool/plant me.
- Cross-pollination: doosre phool me (hawa, paani, insect se).

FERTILISATION: pollen stigma par -> pollen tube niche ovary tak -> male gamete ovule me egg se milta -> ZYGOTE.

Fertilisation ke baad:

- Ovule -> SEED (beej). Ovary -> FRUIT (phal). Zygote -> embryo.

GERMINATION = seed se naya paudha ugna (paani+hawa+temperature theek ho to).

4. Sexual Reproduction in Humans [VERY IMPORTANT]

PUBERTY = bachpan se bade hone ka samay jab reproductive organs kaam shuru karte (ladkiyon me ~10-12, ladkon me ~13-14 saal, alag-alag).

Changes (secondary sexual characters):

- Ladke: daadhi-moochh, awaaz bhari, body+chest hair.
- Ladki: breast develop, menstruation shuru.

MALE reproductive system:

- TESTES (2): sperm + testosterone (hormone) banate; body ke bahar (scrotum) me - kyunki sperm ke liye thanda (body se kam) temperature.
- Vas deferens -> sperm le jaata; prostate + seminal vesicle se ras milta (SEMEN). Urethra se bahar.

FEMALE reproductive system:

- OVARY (2): egg (ovum) + oestrogen banata. Janm se hi egg hote.
- Fallopian tube (oviduct): yahi FERTILISATION hota.
- UTERUS (garbhashay): embryo yahi develop. Vagina se sperm enter.

Fertilisation -> zygote -> uterus ki deewar me chipakta (implantation) -> PLACENTA ke through baby ko ma se oxygen+bhojan milta.

5. Menstruation [IMPORTANT]

Har mahine uterus ki deewar (lining) baby ke liye taiyaar hoti.

Agar egg fertilise NAHI hua -> lining toot kar blood ke saath bahar ->

MENSTRUATION (~28 din ka cycle, 2-8 din chalta).

Matlab: menstruation = pregnancy nahi hui.

6. Reproductive Health (Birth Control + Diseases) [IMPORTANT]

Contraception (garbh-nirodh) ke tareeke:

- Barrier: condom (STD se bhi bachav).
- Hormonal: pills (oral) - hormone balance badalta.
- IUCD (Copper-T) - uterus me lagta.
- Surgical: vasectomy (male) / tubectomy (female).

STDs (Sexually Transmitted Diseases): Gonorrhoea, Syphilis (bacteria);

HIV-AIDS, warts (virus). Condom se bachav.

Female foeticide (kanya bhroon hatya) galat + gairkanooni - sex ratio bigadta.

7. Quick Revision - One Liners

- Asexual: 1 parent, clone, kam variation (fission/budding/fragmentation/regeneration/vegetative/spore).
- Sexual: 2 parent, gamete, jyada variation.
- Flower: stamen (anther+filament male); carpel (stigma+style+ovary female).

- Ovule->seed, Ovary->fruit, zygote->embryo.
- Human: testes (sperm), ovary (egg), fertilisation in fallopian tube, placenta se nutrition; menstruation = no pregnancy.

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - Sexual repro in flowering plant (poora process)

Q: Phool me parpaagan (pollination) se fertilisation aur seed/fruit banne tak ka pura process likho.

Solution:

- Pollination: pollen anther se stigma tak (self ya cross).
- Pollen stigma par chipakta -> POLLEN TUBE banakar style se hote hue ovary tak pahunchta.
- Male gamete (pollen me) + female gamete (ovule me egg) -> FERTILISATION -> ZYGOTE.
- Zygote -> embryo. Ovule -> SEED. Ovary -> FRUIT. Baaki bhaag jhad jaate.
- Seed germinate hokar naya paudha.

Example 2 - Asexual ke types match

Q: Amoeba, Hydra, Plasmodium, Spirogyra, Rhizopus me kaunsa asexual method?

Solution:

- Amoeba: binary fission. Plasmodium: multiple fission.
- Hydra: budding (+ regeneration). Spirogyra: fragmentation.
- Rhizopus: spore formation.

Example 3 - Variation aur sexual reproduction

Q: Sexual reproduction me variation jyada kyun? Iska kya faayda?

Solution:

- Do parents ka DNA mix hota + DNA copying me chhote errors -> har offspring thoda alag (variation).
- Faayda: badalte environment me kuch variants survive kar jaate -> species ko evolution + survival me madad.

Example 4 - Testes body ke bahar kyun

Q: Manav me testes scrotum (body ke bahar) me kyun hote hain?

Solution:

- Sperm banne ke liye body se thoda KAM (thanda) temperature chahiye.
- Isliye testes scrotum me, body ke bahar latke hote hain.

Example 5 (EASIEST) - Phool ke male/female bhaag

Q: Phool ka male aur female bhaag kaunsa hai?

Solution:

- Male = Stamen (anther + filament). Female = Carpel/Pistil (stigma+style+ovary).

PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

- Q1. Reproduction kyun zaroori hai (individual ya species ke liye)?
- Q2. Binary aur multiple fission me antar (example सहित).
- Q3. Phool ke male aur female भाग के नाम.
- Q4. Menstruation क्या बताता है?

MEDIUM (2-3 marks)

- Q5. Asexual reproduction के कौन 4 types + example.
- Q6. Vegetative propagation क्या है + 2 फायदे.
- Q7. Pollination के 2 types में अंतर.
- Q8. Contraception के 3 तरीके (अलग-अलग type).

HARD (3-5 marks, board favourite)

- Q9. Flowering plant में pollination से fruit बनने तक का पूरा process.
- Q10. Human female reproductive system + fertilisation कहां + placenta का role.
- Q11. Sexual repro में variation ज्यादा क्यों + फायदा (evolution से link).
- Q12. Reproductive health: STD + female foeticide पर लिखो.

ANSWER HINTS (PART D)

- A1. Individual के लिये जरूरी नहीं, पर SPECIES के survival के लिये जरूरी.
- A2. Binary = 1 cell 2 में (Amoeba); Multiple = 1 cell बहुत में (Plasmodium).
- A3. Male = stamen; Female = carpel/pistil.
- A4. Egg fertilise नहीं हुआ = pregnancy नहीं (uterus lining बाहर).
- A5. Fission, budding (Hydra), fragmentation (Spirogyra), spore (Rhizopus).
- A6. जड़/तना/पत्ती से नया पौधा; तेजी से + same quality + बिना seed वाले.
- A7. Self (उसी फूल); Cross (दूसरे फूल, हवा/insect से).
- A8. Barrier (condom), hormonal (pills), IUCD/surgical (vasectomy/tubectomy).
- A9. Pollination->pollen tube->fertilisation->zygote; ovule=seed, ovary=fruit.
- A10. Ovary(egg)->fallopian tube (fertilisation)->uterus; placenta से nutrition.
- A11. 2 parents DNA mix + errors -> variation; survival/evolution में मदद.
- A12. STD (gonorrhoea/syphilis/HIV) condom से बचाव; female foeticide गलत.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

Generated by Kiro for Krishna1k
